

Can artificial intelligence support creativity in early design processes?

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Abstract

This study focuses on Generative Artificial Intelligence (AI) and its transformative impact on design ideation. Generative AI, recognized for its ability to produce a wide array of design alternatives, has become an important tool in design, reshaping traditional methodologies. It facilitates the generation of novel and diverse design forms, acting as a co-creator in the design process. This technology, through machine learning and pattern recognition, analyzes extensive design datasets, enabling the production of innovative solutions. The utilization of generative AI extends beyond replicating AI-provided solutions; it aids in developing and influencing novel concepts, thus fostering original design solutions. This aligns with the concept of ‘reflective practice’ in design, where designers iteratively refine concepts through a dialogue between thought and action. The study employed a quasi-experimental design with 40 design students, randomly assigned to two groups of 20 each. Conducted in two phases, each phase involved a distinct urban furniture design task. In Phase 1, Group A was provided with a text-to-image generating AI tool, while Group B was not. In Phase 2, both groups undertook a similar task without AI assistance. This design exercise allowed for examining the influence of AI on creativity and cognitive load. Design outcomes from both tasks were anonymized and evaluated by experienced professionals using the Creative Product Semantic Scale (CPSS), which measures Novelty, Resolution, and Elaboration and Synthesis. Additionally, the NASA Task Load Index (NASA TLX) questionnaire assessed cognitive load aspects such as mental demand and effort. Findings suggest that generative AI significantly influences the creative design process, enhancing the quality of design outcomes and reducing cognitive load. The AI group demonstrated better performance in both tasks, indicating the impact of AI tools on design skills. This study underscores the potential of AI tools in design education, balancing cognitive load management with creativity enhancement.

Keywords

Artificial intelligence, design ideation, cognitive load, creative process, design process

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Introduction

For over six decades, questions like “Can machines think?” and “Can machines be intelligent?” have been subjects of ongoing debate. Allan Turing¹ developed his infamous Turing test that has been used as a way of judging the ability of a machine to understand. Subsequent commentaries to the Turing test have suggested that machines can never duplicate human thought processes and no matter how well machines imitate human behavior, they will never truly understand in the same way as a human being.²

As AI (Artificial Intelligence) becomes increasingly integrated with day-to-day human activities it is important to reevaluate how we as humans understand and cohabitate with AI in the future. AI has the potential to automate mundane tasks that would potentially relieve humans to focus on more creative aspects. However, with the rise of Creative AI, the traditional boundaries of AI automation are blurring. De Vries³ proposes three types of use cases for creative AI, (1.) Understanding (since creative behavior requires some form of understanding). (2.) Representation (using synthetic data to fill in missing data) and (3.) Creations (such as transforming images to create novel products, and text-to-image translations). With these new developments in technology, in addition to questioning the creativity of AI, it is important to question if creative AI can help designers be more creative?

Given the potential of creative AI, the Integration of AI in early design education has been studied and discussed by several researchers.^{4,5} However, these studies have not focused on how AI might affect creativity in the design process, especially with students in early design studios. Can we consider AI as just another digital technology like many others? Or will it create a paradigm shift in the way that we design and think about creativity and design?

Literature review

Generative AI and design

In order to understand how AI has impacted and been perceived in design education and the industry, it is important to provide a clear definition of Generative AI. Generative AI, a subset of AI, includes machines creating content, data, or outputs without explicit human programming, utilizing generative models rooted in learned patterns from existing data. The integration of generative AI in design processes has demonstrated numerous advantages, including elevated design project quality, improved workflow efficiency, and streamlined construction tasks.⁶ Researchers, have also introduced a cost-benefit index to gauge the impact of automation and AI adoption in the Architecture, and Construction sector. This index not only acknowledges the replacement of manual jobs but extends its implications to roles requiring analytical, intuitive, and empathetic skills.⁷ Furthermore, studies have underscored the positive influence of AI on the construction industry, contributing to increased productivity and overall output, positioning AI as a crucial component for economic stability.⁸

However, there is a critical need to understand the potential drawbacks associated with AI in design. Concerns include system downtime, programming errors, and misinterpretations of regulations, urging a cautious approach despite the evident advantages.⁹ A separate study involving 140 instructors across diverse programs acknowledged the personalized efforts exerted by human teachers for individual students, tailoring their approach based on each student's unique knowledge and skills. This personalized engagement is considered crucial for achieving success in the learning process, a feat that artificial intelligence (AI) may struggle to replicate. While AI can suggest design ideas to students, there remains a gap as students may implement these suggestions without possessing sufficient knowledge to critically assess the work.¹⁰ In light of this, the challenges of incorporating artificial intelligence systems in educational settings become evident. Another study highlights the associated costs of implementing AI and underscores the significance of human

relationships in the classroom, emphasizing the irreplaceable role of human educators in fostering a comprehensive and nuanced learning experience.¹¹

As the world steadily transitions towards increased AI adoption, the literature review emphasizes the imperative to equip students with the requisite skills. This ensures that the next generation of professionals is adept at effectively utilizing AI-facilitating tools, positioning them as key contributors to the ever-evolving landscape of design and construction.

In this regard, generative AI also represents a significant advancement in the field of design ideation. Its ability to generate diverse design options, enhance collaboration contribute to the overall creative design process. Generative AI is increasingly recognized as a pivotal tool in design, offering innovative approaches to conceptualization and ideation. This technology, which encompasses algorithms capable of generating new design forms and solutions, is reshaping the traditional methodologies of design. A key advantage of generative AI in design is its capacity for creating a diverse range of design alternatives. This is particularly beneficial in the early stages of design, where exploring various possibilities is crucial. Generative AI algorithms, through machine learning and pattern recognition, can analyze extensive datasets of designs, enabling them to produce novel and often unorthodox design solutions. This aspect is highlighted in the work of Janssen and Stouffs,¹² who discuss how generative AI can act as a co-creator in the design process, offering a multitude of design options that might not be immediately apparent to human designers.

Generative AI also plays a significant role in optimizing designs for various parameters such as structural integrity, material efficiency, and environmental sustainability. For example, Ko et al.¹³ demonstrate how AI algorithms can be used to optimize building layouts and structures for better environmental performance, thus contributing to sustainable design practices. Furthermore, the integration of generative AI in design aligns with the increasing use of Virtual Reality (VR) and Augmented Reality (AR) technologies. These technologies, when combined with AI, can lead to more immersive and interactive design experiences. Kensek¹⁴ explores how the amalgamation of generative AI with VR/AR technologies can enhance the design process, allowing architects to visualize and interact with AI-generated designs in real-time virtual environments.

Lateral transformations in the design process represent a paradigm shift towards embracing alternative, often unconventional, sources of inspiration and ideation. The influence of external stimuli plays a crucial role in shaping design outcomes. This concept is explored in depth in the study by Chandrasekera et al.,¹⁵ which investigates the impact of subliminal stimuli on the design process, demonstrating how subtle external influences can significantly alter design perceptions and outcomes. The integration of Generative AI in the design process presents a novel approach to harnessing external stimuli for creative ideation. AI-generated images, in particular, offer visual stimuli may lead to innovative design solutions. These images, derived from complex algorithms, often contain patterns, forms, and compositions that are not typically encountered in traditional design processes. As noted by Janssen and Stouffs,¹² the use of generative AI in design can introduce a level of complexity and novelty that significantly deviates from conventional design norms, thereby fostering creativity.

The potential of AI-generated images to influence design ideation lies in their ability to present visual cues that are both unexpected and inspiring. These cues can trigger lateral thinking, a cognitive process essential for innovative design. Lateral thinking in design, as discussed by De Bono,¹⁶ involves the search for solutions through an indirect and creative approach, using reasoning that is not immediately obvious. AI-generated images, with their inherent unpredictability and richness, serve as a perfect catalyst for such thinking. Furthermore, the use of AI in generating design stimuli aligns well with the concept of 'design by analogy', a method where designers draw parallels and inspiration from seemingly unrelated fields or phenomena. This approach is highlighted in the work of Goel,¹⁷ who emphasizes the importance of analogical reasoning in creative design. To explore creativity demonstrated by AI, one notable instance is AlphaGo, an AI developed by Google's DeepMind, which made history by defeating Lee Sedol, one of the world's best Go players. This event highlighted AI's ability to not only master complex games but also to exhibit creative strategies that

surprised the human champion. Such achievements underscore the potential of AI in contributing creatively to areas previously dominated by humans, raising questions and possibilities about AI's role in creative processes.

Creativity

This paper aims to closely examine how AI's ability to generate new ideas impacts designers' creativity, especially during the early stages of coming up with design concepts, starting by exploring the complex nature of creativity in design, and then looking at its different aspects and types. This approach helps us understand the role of AI in enhancing creativity and its importance in the initial steps of designing.

The term "creativity" has been used – or more precisely misused – especially in the context of design and design education. Despite Torrance's¹⁸ statement that "creativity defies precise definition," many definitions of creativity have been provided. Discussions and commentaries on creativity suggest that the definitions provided may be categorized into multiple groups. Rhodes¹⁹ theorized that creativity falls into four distinct categories: individual aspects (the individual who is involved in the creative process), cognitive aspects (the creative process), the influence of the context where the creative process is taking place (place), and the resulting creative product. These are known as the four P's of creativity (person, process, place/press, and product).

As suggested by Rhodes¹⁹ creativity can be primarily discussed either in terms of the product or in terms of the process.²⁰ Rosenman and Gero stated that a product can be considered to be the result of a creative process depending on the innovativeness of the product and the value and richness of interpretation and that the creativity of the process can be described from an information processing standpoint. They discussed the creative aspects of the creative process through entropy, efficiency, and richness. The aspect of novelty has been central to a number of definitions of creativity.²¹ Hausman²² stated that "each appearance of genuine novelty is a sign of creative activity" (p. 20). However, these definitions of creativity and its connection with novelty have been much debated.

Gero and Maher²³ focused on the creative design process and stated that creativity introduces new variables to the design process which were not originally considered by the designer or design system; design and creative design are different because reasoning plays a major role in creative design. The term "design creativity" has been widely used.^{24–26} While the Vitruvian virtues of architecture²⁷ are *utilitas* (function), *firmitas* (solidity/stability), and *venustas* (delight/aesthetics), design seldom stops at aesthetics; it goes beyond mere aesthetics and becomes an artifact that makes people think: from aesthetics to mindfulness. Taking this into consideration, design creativity cannot only focus on novelty but must also focus on utility and value.²³

The question of AI's creativity hinges on the definition of creativity itself. If we define creativity as the ability to produce work that is both novel and valuable, AI has certainly shown promise in various domains. From composing music to producing artwork and even writing poetry, AI algorithms have created outputs that, to some observers, may be indistinguishable from those produced by humans. AI's ability to sift through vast datasets and identify patterns can lead to the generation of unique combinations of ideas, which is a key component of what we traditionally consider creative work. However, AI's "creativity" is fundamentally different from human creativity. AI lacks consciousness and intentionality (for the timebeing)—it does not have desires, emotions, or experiences and does not create with the intention to express itself or convey emotion. Instead, AI operates within the parameters set by its programming and the data it has been fed. Its "creativity" is the result of complex calculations and probability estimations rather than a conscious desire to create something new or meaningful. AI also lacks the ability to contextualize its creations within the broader human culture and history, which is often vital to what we consider true creative acts. Moreover, AI's reliance on existing data to generate output can lead to a replication of what has already been done, rather than truly innovative leaps. Its creative process is more about recombining known elements in new ways based on

algorithms and patterns it has learned. While this can result in novel combinations, it is arguable whether this constitutes creativity in the human sense.

However what if the same method of using patterns and combinations to develop novel product can be combined with the human emotions and contextualization to essentially teach humans to look at patterns where they are not readily visible through AI, in almost a way of reverse teaching (as AI is trained through human data and then if humans can be trained on the AI patterns), it would be interesting to see this application in design and design education.

Product-focused assessment of an artifact is particularly relevant in fields like architecture and design, where the functional and aesthetic qualities of the final product are important. Furthermore, Dorst²⁸ emphasizes that focusing on product creativity allows for a more direct correlation between the creative output and its impact on users and society. This perspective is crucial in disciplines like interior design and architecture, where the end product interacts with and influences human behavior and social dynamics. Therefore, in this study the creativity in the product is assessed through established scales. The question with regard to the topic of this study is not if AI is truly creative or not, it is about how can Creative AI help design students understand creativity.

In this study the main research questions are as follows.

RQ1: “In what ways does the Cognitive Load vary between design tasks completed with and without the assistance of text-to-image generating AI?”

This question aims to explore the specific aspects of cognitive load that are affected by the use of AI in the design process.

RQ2: “What are the differences in creativity, as measured by Novelty, Resolution, and Elaboration and Synthesis, between design outcomes produced with AI assistance and those produced without it?”

This question is structured to directly address the dimensions of creativity defined by the Creative Product Semantic Scale (CPSS).

Design concepts

In most of the studies that have been conducted on design and the design process, the evolution of the design process is discussed as a step-by-step process. How does a designer begin to design? Does the designer begin with arbitrary sketching? Lawson²⁹ provided two contrasting styles of operation: problem focused and solution focused. He stated that in solving design problems, science students use a problem focus, which is much more analytical, while design students use a solution focused approach in which they try out different solutions and see what goes wrong. Hillier et al.,³⁰ who provides a conjecture-analysis model for the design process, states that in order to make a problem tractable, it should be pre-structured, either explicitly or implicitly. He further states that design is essentially a matter of pre-structuring problems and argues that this is the reason that design is resistant to empirical rationality, where even with a complete account of the designer's operations, there will still be gaps as to where the solution originated. Darke³¹ modifies Hillier's model as generator-conjecture-analysis. She rationalizes this model by stating that the idea of a primary generator precedes the conjecture stage. She defines the primary generator as the concept or objective that generates a solution. Given the importance of this conceptual stage in the design process, there is a critical need to understand how AI might affect concept generation. Especially in early design studios, when students

are beginning to learn how design concepts are formulated, could AI assist in teaching students about abstraction?

The centrality of design concepts in the design process cannot be overstated; they are the bedrock upon which all subsequent design decisions and developments are built. These concepts serve not only as the initial spark of creativity but also as the guiding framework that shapes the trajectory of the entire design process. The genesis and evolution of design concepts are deeply influenced by a myriad of factors, including inspiration from various stimuli and the transformative process of lateral thinking. Inspiration for design concepts often emerges from diverse and sometimes unexpected sources. These can range from the tangible – such as the natural environment, urban landscapes, and historical contexts – to the intangible, like cultural narratives, personal experiences, or theoretical discourses. This eclectic pool of stimuli provides a valuable resource for ideas and perspectives that designers draw upon to conceive and develop their design concepts. As Lawson²⁹ notes, the best design concepts often arise from a deep understanding and synthesis of multiple influences, leading to solutions that are both innovative and contextually relevant.

Lateral transformation plays a pivotal role in the development of design concepts. This approach encourages designers to step beyond traditional boundaries and explore a broader range of possibilities. Lateral thinking in design, as advocated by De Bono,¹⁶ involves seeking solutions through creative and divergent thinking, often leading to more innovative and unique design outcomes. In the context of design education, particularly for students, the development of design concepts often involves a process of abstraction. Abstraction in design education is a method through which students learn to distill complex realities into simpler, more fundamental forms and ideas. This process is crucial in developing the ability to conceptualize and articulate design ideas effectively. As Schön³² discusses, the process of abstraction allows design students to engage with the ‘problematic’ aspects of a design situation, enabling them to develop a deeper understanding and more nuanced design responses. Through abstraction, design students learn to navigate the balance between theoretical concepts and practical considerations, developing a critical skill set that allows them to conceptualize and communicate their ideas effectively. This skill is essential in the design process, where the ability to abstract complex ideas into coherent design concepts is fundamental.

The development of design concepts is a critical phase in the design process, shaped significantly by inspiration, lateral transformation, and the process of abstraction, especially in the context of design education. These concepts not only guide the design process but also embody the creative and intellectual rigor of the designer. As such, the nurturing of these skills in design students is paramount, ensuring the continual evolution and enrichment of the design profession. Generative AI, particularly through the use of AI-generated images, offers a novel approach to facilitating abstraction and thereby enhancing the lateral thinking process in design students. Generative AI, with its capability to produce complex and often unexpected visual outputs, provides a unique platform for students to engage in abstraction. This process of engaging with AI-generated images can stimulate the cognitive mechanisms underlying lateral thinking, as students are prompted to interpret, deconstruct, and recontextualize these images in the pursuit of innovative design concepts. The role of such technology in fostering creative thinking is emphasized by Oxman³³ who highlights how computational design tools can expand the creative range of designers, enabling the exploration of new forms and patterns that might not emerge through traditional design methods. By integrating generative AI into the design curriculum, educators can provide students with a unique tool that not only aids in the development of abstraction skills but also catalyzes the lateral thinking process, thereby enriching the overall process of design concept development.

The utilization of Generative Artificial Intelligence (AI) in the design process extends far beyond the mere replication of AI-provided solutions; it fundamentally aids in the development and influence of novel concepts, fostering the creation of innovative and original design solutions. Generative AI, through its advanced algorithms, generates countless design possibilities that serve not as final solutions but as catalysts for creative thinking. The true value of generative AI lies in its ability to expand the designer’s creative

horizon, challenging conventional design paradigms and encouraging exploration into uncharted territories of design possibilities. As Janssen and Stouffs¹² state, generative AI acts as a co-creator in the design process, offering novel perspectives that can significantly alter and enrich the conceptual development phase. Furthermore, the interaction with AI-generated designs compels designers to critically assess and reinterpret these suggestions, leading to a deeper understanding of design principles and the cultivation of original ideas. This process aligns with the concept of 'reflective practice' in design, as described by Schön³² where designers learn from each iterative step, refining their concepts through a continuous dialogue between thought and action. In essence, generative AI serves as a powerful tool in the designer's tool palette, not as a source of ready-made solutions, but as a springboard for innovation and the development of truly novel design concepts.

Cognitive load and design creativity

The relationship between cognitive load and efficiency in task performance can be conceptualized as a bell curve, illustrating a nuanced interplay where efficiency initially increases with cognitive load, reaches an optimal point, and then decreases as the load continues to rise. This phenomenon can be understood through the lens of cognitive load theory, which posits that working memory has a limited capacity for processing information.³⁴ The concept of 'desirable difficulties', as proposed by Bjork,³⁵ supports this idea, suggesting that certain levels of challenge in cognitive tasks can enhance learning and performance. As cognitive load continues to increase, individuals reach a point of optimal efficiency – the peak of the bell curve. This peak represents a state where the individual is effectively managing the cognitive demands of the task, leading to maximum efficiency. At this stage, the balance between task complexity and cognitive capacity is ideal, allowing for optimal performance. This state is often characterized by a state of 'flow', as described by Csikszentmihalyi,³⁶ where individuals experience deep immersion and focus in their activities. However, beyond this optimal point, as cognitive load continues to increase, efficiency begins to decline. This decline is due to the overload of working memory, where the amount of information and the complexity of tasks exceed the individual's cognitive processing capacity. This phenomenon is aligned with the findings of Kalyuga et al.,³⁷ who demonstrated that when cognitive load surpasses an individual's ability to process information, performance and efficiency deteriorate.

The relationship between cognitive load and creativity in design tasks can also be conceptualized in the same way. Cognitive load, defined as the total amount of mental effort being used in the working memory, can significantly influence the creative output in design tasks. High cognitive load is often thought to impede creativity, as it can restrict the mental space necessary for divergent thinking, a key component of creative ideation. Kershaw et al.³⁸ research suggests that reducing cognitive load through structured interventions can lead to greater creativity in engineering design by facilitating divergent thinking. Divergent thinking requires the ability to generate multiple, novel ideas, which can be hampered when the working memory is overloaded with information or complex problem-solving tasks. Therefore, managing cognitive load is crucial for facilitating divergent thinking in the design process. By understanding and mitigating the factors that contribute to cognitive overload, designers can create an environment conducive to innovative and creative thinking.

Conversely, a moderate level of cognitive load can sometimes enhance creativity. This is aligned with the concept of 'desirable difficulties' proposed by Bjork,³⁵ suggesting that certain challenges in learning and problem-solving can actually improve long-term performance and creativity. In the context of design, a balanced cognitive load can stimulate creative thinking by encouraging designers to seek novel solutions and think outside conventional frameworks. Furthermore, the nature of the design task itself can influence the relationship between cognitive load and creativity. Complex, open-ended design tasks may require a high level of cognitive engagement, which, if managed effectively, can lead to highly creative outcomes. This is highlighted in the work of Cross,³⁹ who emphasizes the importance of expertise and experience in managing

cognitive load during complex design tasks. Therefore, the impact of cognitive load on creativity in design tasks is multifaceted. While high cognitive load can potentially constrain creative thinking, a balanced or moderate cognitive load can foster creativity by introducing challenges that stimulate innovative problem-solving. The effectiveness of managing cognitive load in creative tasks is also influenced by the designer's expertise and the nature of the design task, underscoring the importance of developing strategies to optimize cognitive resources during the design process. In this project, we meticulously crafted a design task calibrated to induce an optimal level of cognitive load, thereby enabling the researchers to effectively measure and closely observe the nuances of the creative design process.

Method

The primary objective of this study was to investigate the impact of Creative AI on the design process, focusing specifically on the concept generation phase. The study aimed to explore how the integration of AI influences creativity and cognitive load in the design process. A total of 40 students from early design studios were recruited for this study. The participants were selected based on specific criteria, including their year of study, prior experience with design tasks, and familiarity with design software.

The study employed a quasi-experimental design, with participants randomly assigned to two groups. Each group consisted of 20 students. The study was conducted in two phases, each involving a distinct design task. In Phase 1 (Urban Furniture Design Task 1) both groups were given an identical design brief to create a piece of urban furniture. The brief included specific requirements regarding dimensions, materials, and intended use. Group A (the experimental group) was provided access to a text-to-image generating AI tool to assist in the conceptualization phase. Group B (the control group) did not receive AI assistance. Participants were instructed to focus on concept development, documenting their process through sketches and 3D renderings. In Phase 2 (Urban Furniture Design Task 2) Upon completion of the first task, both groups were assigned a second, similar design brief for another piece of urban furniture. For this task, neither group was permitted to use AI assistance, ensuring a consistent condition for comparison. Both tasks were intentionally designed to be similar, providing students with the opportunity to apply and refine their learnings from the first task in the subsequent one.

The participants' design outcomes were collected and anonymized for evaluation. A total of 80 designs (40 from each task) were reviewed by two experienced design professionals. The reviewers were blinded to the group assignments and the use of AI in the design process. The Creative Product Semantic Scale (CPSS), based on the theoretical model by O'Quin & Besemer⁴⁰ was utilized to evaluate the creativity of the final designs. This scale measures three dimensions of product creativity: Novelty, Resolution, and Elaboration and Synthesis. Each design was scored independently by the reviewers, and the scores were averaged for analysis. To assess the cognitive load experienced by participants in both conditions, the NASA Task Load Index (NASA TLX) questionnaire was administered after each design task. This self-report measure evaluates various aspects of cognitive load, including mental demand, effort, and frustration.

Creative Product Semantic Scale (CPSS) is used for measuring creativity in this project given its robust framework and relevance in the context of cognitive load. The CPSS offers a multidimensional approach to assessing creativity, focusing on Novelty, Resolution, and Elaboration and Synthesis. This comprehensive model is highly suitable for evaluating the creative outcomes of design tasks, especially in studies where cognitive load is a significant variable. Novelty dimension of the CPSS assesses the originality and uniqueness of the design outcomes. In the context of this project, where cognitive load is carefully modulated, the measurement of Novelty provides insights into how cognitive constraints or enhancements influence the generation of original ideas. The Resolution dimension evaluates the usefulness and practicality of the creative product, including its problem-solving effectiveness. This is particularly relevant in design tasks where cognitive load might impact the decision-making and problem-solving abilities of the participants. The

ability to create designs that are not only novel but also practical and solution-oriented is a key indicator of creativity under varying cognitive loads.³⁷ Elaboration and Synthesis assess the degree of detail and the integration of elements within the design. This dimension is critical in understanding how cognitive load influences the complexity and refinement of design outcomes. It aligns with the concept that moderate cognitive load can enhance the depth and elaboration of creative work.³⁵ Therefore, CPSS provides a nuanced and multidimensional approach to assessing creativity, making it an ideal tool for this project. Its ability to evaluate various aspects of creative products offers valuable insights into the interplay between cognitive load and creativity in the design process. The scale's comprehensive nature ensures that the creativity assessment is not only focused on the originality of ideas but also encompasses their practicality and the intricacy of their execution, which are essential components in understanding the full spectrum of creativity in design tasks. The NASA Task Load Index (NASA-TLX) is a widely recognized tool for assessing cognitive load, particularly in the context of the design process. Developed by Hart and Staveland,⁴¹ the NASA-TLX is a multidimensional scale designed to obtain workload estimates from operators based on six subscales: mental demand, physical demand, temporal demand, performance, effort, and frustration. Its application in design processes is crucial for understanding the cognitive demands placed on designers and for optimizing design tasks to enhance efficiency and creativity.

The study was conducted in accordance with ethical guidelines for research with human subjects and was provided Institutional Review Board (IRB) approval. Participants were informed about the study's purpose, procedures, and their right to withdraw at any time without penalty. Informed consent was obtained from all participants prior to the commencement of the study.

Analysis and discussion

Assessment of cognitive load using NASA TLX

An independent samples *t* test was conducted to compare Task 1 and Task 2 scores between two groups, AI (coded as 1) and Non-AI (coded as 0). For Task 1, the AI group ($N = 20$) had a mean score of 3.06 (SD = 0.63), and the Non-AI group ($N = 20$) had a mean score of 3.45 (SD = 0.59). The test indicated a significant difference between the two groups, $t(38) = -2.035$, $p = .049$. The mean difference was -0.39 (95% CI: -0.78 to -0.002), suggesting that the cognitive load in the Non-AI group was significantly higher than the AI group on Task 1 (Tables 1–6).

For Task 2, the AI group ($N = 20$) had a mean score of 3.05 (SD = 0.84), while the Non-AI group ($N = 20$) had a mean score of 3.58 (SD = 0.75). The independent samples *t* test showed a significant difference, $t(38) = -2.097$, $p = .043$. The mean difference was -0.53 (95% CI: -1.04 to -0.02), indicating that the cognitive load in the Non-AI group was significantly higher than the AI group on Task 2.

These results suggest that for both Task 1 and Task 2, participants in the Non-AI group were effected by a higher cognitive demand. The differences were statistically significant with small to moderate effect sizes, as indicated by the mean differences and confidence intervals.

The significant differences in cognitive load in the two groups for both Task 1 and 2 offer insightful contributions to the understanding of cognitive load and its impact on creativity in design tasks, particularly in

Table 1. Task 1 group statistics for cognitive load.

Group (AI or Non-AI)	N	Mean	Std. Deviation	Std. Error mean
AI (1)	20	3.058	.631	.141
Non-AI (0)	20	3.450	.585	.131

Table 2. Independent Samples Test Table for Task 1 for cognitive load.

Variance equality	F	Sig	t	df	Sig. (2-Tailed)	Mean difference	Std. Error difference	95% CI lower	95% CI upper
Equal variances assumed	.265	.610	−2.035	38	.049	−.392	.192	−.781	−.002
Equal variances not assumed			−2.035	37.78	.049	−.392	.192	−.781	−.002

Table 3. Task 2 group statistics for cognitive load.

Group (AI or Non-AI)	N	Mean	Std. Deviation	Std. Error mean
AI (1)	20	3.051	.839	.188
Non-AI (0)	20	3.578	.747	.1678

Table 4. Independent Samples Test Table for Task 2 for cognitive load.

Variance equality	F	Sig	t	df	Sig. (2-Tailed)	Mean difference	Std. Error difference	95% CI lower	95% CI upper
Equal variances assumed	1.245	.272	−2.097	38	.043	−.527	.251	−1.03	−.018
Equal variances not assumed			−2.097	37.503	.043	−.527	.251	−1.035	−.0179

Table 5. Group statistics for task 1 and task 2 scores for creativity.

Group (AI or NoAI)	Condition	N	Mean	Std. Deviation	Std. Error mean
AI (1)	Task 1	20	.516	.101	.023
Non-AI (0)	Task 1	20	.435	.105	.024
AI (1)	Task 2	20	.545	.102	.023
Non-AI (0)	Task 2	20	.473	.091	.020

the context of using AI-assisted tools. The results indicate that participants in the Non-AI group experienced a higher cognitive demand in both design tasks, with statistically significant differences exhibiting small to moderate effect sizes. This outcome aligns with the cognitive load theory, which posits that the mental effort required to process information can influence task performance.³⁴

The increased cognitive load in the Non-AI group could be attributed to the absence of AI assistance, which necessitated more intensive mental effort for concept development. This finding is consistent with the research by Kalyuga et al.,³⁷ who noted that the absence of appropriate instructional support could increase cognitive load and potentially hinder performance. In contrast, the AI-assisted group (Group A) likely benefited from the cognitive offloading provided by the AI tool, which could have facilitated a more efficient allocation of cognitive resources, as suggested by Fiorella and Mayer⁴² in their work on the role of external aids in reducing cognitive load. Interestingly, the similarity in design tasks across both phases provided an

Table 6. Independent samples test for task 1 and task two for creativity.

Condition	Variance equality	F	Sig	t	df	Sig. (2-Tailed)	Mean difference	Std. Error difference	95% CI of the difference lower	95% CI of the difference upper
Task 1	Equal variances assumed	1.602	.213	2.461	38	.018	.080	.033	.014	.146
	Equal variances not assumed			2.461	37.930	.019	.080	.033	.014	.146
Task 2	Equal variances assumed	.099	.755	2.370	38	.023	.07239	.03055	.01055	.13423
	Equal variances not assumed			2.370	37.541	.023	.07239	.03055	.01053	.13425

opportunity for participants to apply learnings from the first task to the second. However, the continued higher cognitive load in the Non-AI group during the second task suggests that the absence of AI support consistently imposed greater cognitive demands, regardless of prior experience with a similar task. This observation underscores the potential of AI tools in reducing cognitive load and enhancing learning transfer, as discussed by Van Merriënboer and Sweller⁴³ in their exploration of cognitive load theory in educational design.

Assessment of creativity using CPSS

An independent samples *t* test was conducted to examine differences in scores between AI (coded as 1) and Non-AI (coded as 0) groups across two conditions: Task 1 and Task 2. In Task 1, the AI group ($N = 20$) had a mean score of 0.516 ($SD = 0.101$), while the Non-AI group ($N = 20$) had a mean score of 0.435 ($SD = 0.105$). The *t* test revealed a significant difference between the groups, $t(38) = 2.461$, $p = .018$. The mean difference was 0.080 (95% CI: 0.014 to 0.146), indicating that the AI group had significantly higher scores compared to the Non-AI group in the Task 1 condition.

In the Task 2, the AI group ($N = 20$) had a mean score of 0.545 ($SD = 0.102$), and the Non-AI group ($N = 20$) had a mean score of 0.473 ($SD = 0.091$). The results showed a significant difference, $t(38) = 2.370$, $p = .023$. The mean difference was 0.072 (95% CI: 0.011 to 0.134), suggesting that the AI group also scored significantly higher than the Non-AI group in Task 2.

These findings suggest that the AI group consistently outperformed the Non-AI group in both Task 1 and Task 2, with statistically significant differences in scores. The mean differences and confidence intervals indicate a consistent pattern of higher performance by the AI group. However the difference in mean scores in AI and Non AI group is higher in Task 1, since the students presented work that used AI software to generate the solution that were reviewed (Figures 1 and 2).

The results of the independent samples *t* test provide compelling evidence regarding the impact of AI assistance on design task performance. The significant differences in mean scores between the AI and Non-AI groups in both Task 1 and Task 2 offer insights into the efficacy of AI tools in enhancing design outcomes. In Task 1, the AI group's higher mean score suggests that the integration of AI in the design process positively

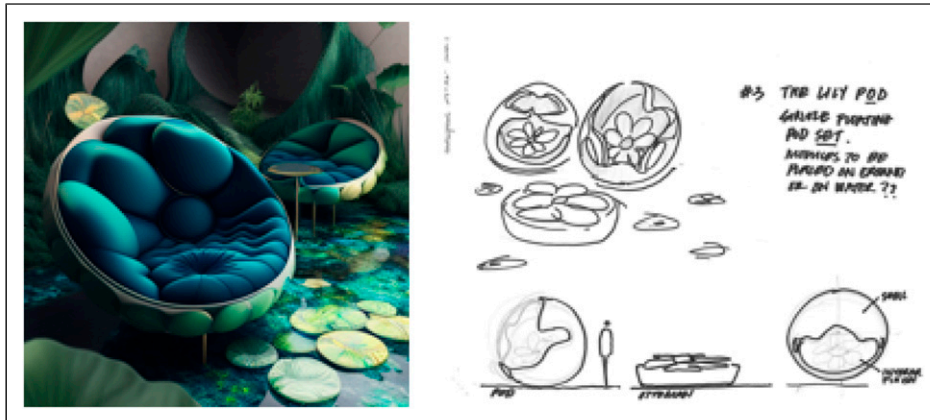


Figure 1. Task 1, AI Group – Example of one of the students AI generated images (on the left) and the resultant design sketch (on the right). The conceptual statement for this generated by students through ChatGPT: The organic seating design draws inspiration from lily pads and floating leaves, with curved lines and customizable upholstery in shades of green. The pods have covered shading and can be personalized using various colors, prints, and textures, creating a natural and calming environment in urban settings.

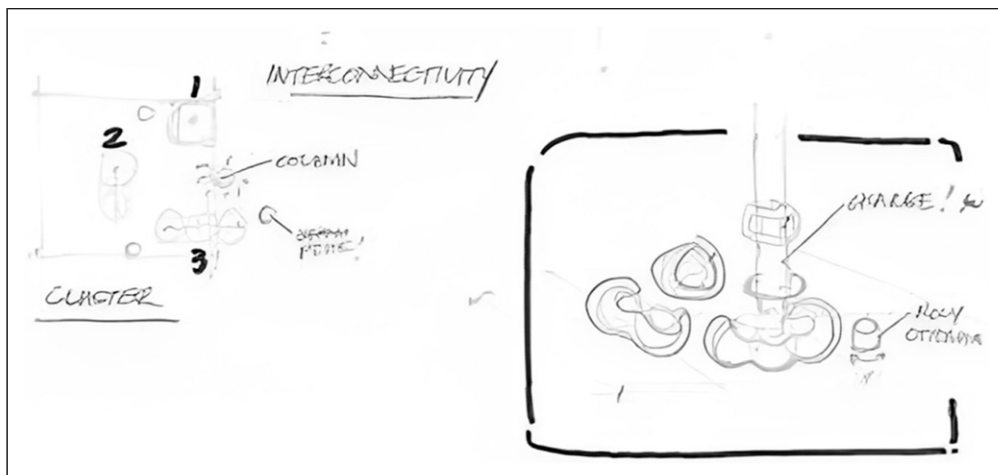


Figure 2. Task 1 Non-AI Group Example of one of the students design sketches.

influenced the quality of the design outcomes as perceived by the reviewers. This finding aligns with the notion that AI tools can augment human creativity by providing novel perspectives and solutions, as suggested by Deterding,⁴⁴ who discusses the enhancement of human performance through digital tools. The AI's capacity to generate diverse and innovative ideas likely contributed to the enhanced performance observed in the AI group.

The continuation of this trend in Task 2, where neither group had access to AI assistance, is particularly intriguing. The AI group's sustained higher performance could be indicative of a residual learning effect, where the exposure to AI-generated concepts in Task 1 enriched the students' creative thinking and problem-

solving skills, subsequently benefiting their performance in Task 2. This phenomenon resonates with the concept of ‘transfer of learning’, as explored by Perkins and Salomon,⁴⁵ who emphasize the importance of prior learning experiences in influencing subsequent task performance.

However, the observed decrease in the mean score difference between the AI and Non-AI groups from Task 1 to Task 2 warrants attention. This reduction could suggest that while AI tools provide an initial boost in creative output, their long-term impact on learning and creativity may be more nuanced. This interpretation aligns with the findings of Mayer,⁴⁶ who cautions against over-reliance on digital tools for cognitive tasks, advocating for a balanced approach that fosters deep learning and understanding.

The study’s findings underscore the potential of AI tools in enhancing design task performance, particularly in the initial stages of exposure. The residual benefits observed in subsequent tasks without AI assistance highlight the importance of integrating such tools in a manner that supports long-term learning and creativity. These insights have significant implications for the incorporation of AI in design education and practice, emphasizing the need to strike a balance between immediate performance enhancement and the cultivation of enduring creative skills.

Conclusion

This study highlights the significant impact of Generative Artificial Intelligence (AI) in enhancing creativity and reducing cognitive load in design processes. The research, focused on early design education, demonstrates how AI tools can serve as effective co-creators, facilitating novel design solutions and fostering skill development in design students. The utilization of AI in the creative process not only aids in generating diverse design options but also instills a deeper understanding of design principles. The results underline the potential of integrating AI into the design curriculum, emphasizing its role in balancing cognitive load and stimulating innovation. This integration can enrich the design process, providing a platform for students to engage in more abstract and lateral thinking. Overall, the study underscores the transformative potential of AI in reshaping traditional design methodologies, suggesting a paradigm shift in the way design education and practice are approached.

The findings from our study highlight several shortcomings in the study as well as areas for future exploration, particularly the impact of individual learning styles on the utilization of generative AI tools in design. These tools, which vary in their approach to generating imagery, underscore the diverse methods available for creative expression. One prevalent technique involves submitting textual prompts to generate visuals. Alternatively, some platforms allow users to input an image and then refine the generated output through sliders that adjust specific features. This diversity in input methods suggests the possibility of future innovations in AI interaction, potentially accommodating a broader range of designer skills and preferences. This exploration is reminiscent of Maya Lin’s compelling written statement for the Vietnam Veterans Memorial, which significantly contributed to the jury’s comprehension of her design’s intended experiential and symbolic impact—a depth that may not have been fully conveyed through drawings and models alone. Similarly, the evolving interface and input methods for generative AI tools highlight the importance of aligning these technologies with the designer’s individual skills and creative processes, thereby enhancing the expressive and conceptual clarity of their work.

Additionally, it’s important to acknowledge one of the study’s main limitations: the focus on a small cohort of 40 students from early design studios who are not experts. Future studies should expand on this foundation by including a more extensive and diverse participant pool, particularly experts from the design industry, to validate and broaden the generalizability of the findings.

However, the potential over-reliance on AI poses a risk to individual creativity and raises concerns about equitable access to AI tools. Future research should explore AI’s long-term impact on design thinking and creativity, alongside the development of ethical frameworks. Interdisciplinary collaborations will be vital in

optimizing AI's role, ensuring a sustainable, ethical, and innovative future in design. The implications of AI in design education include the need for curriculum development that encompasses both design principles and the technical aspects of AI. There's an emphasis on enhancing skills for AI management and fostering creative collaboration with AI systems. Training for cognitive load management is also crucial, balancing AI's efficiency with the need for critical thinking. In the design profession, adapting to AI as a co-creative partner is essential, reshaping roles and navigating ethical considerations like authorship and data privacy.

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